

Diopter is the unit of refractive power of a lens. The power of a lens is defined as the reciprocal of its focal length in metres.

$$D = \frac{1}{f}$$

Where, 'D' is power of lens in diopter and 'f' is the focal length in metres. Converging lens has positive optical power and diverging lens has negative optical power.

41. If the power of a lens glass is +2 diopter, its focal length will be :

- (a) 200 cm (b) 100 cm
(c) 50 cm (d) 2 cm

41st B.P.S.C. (Pre) 1996

Ans. (c)

We know that,

$$\left[D = \frac{1}{f} \right]$$

$$1 \text{ metre} = 100 \text{ cm}$$

$$\text{or } f = \frac{100}{D} \text{ cm}$$

$$f = \frac{100}{2} = 50 \text{ cm}$$

42. The focal length of a convex lens is $f_L = 10$ cm. On immersing in water, it will act as a

- (a) convex lens of $f_L = 10$ cm
(b) concave lens of $f_L = 10$ cm
(c) convex lens of $f_L > 10$ cm
(d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (c)

Given that,

focal length of convex lens, $f_L = 10$ cm.

When this lens immerse in water, it will act as a convex lens of focal length greater than 10 cm. because the refractive index of water is greater than air.

As we know,

$$\text{Focal length of lens, } \frac{1}{F} = (\mu - \mu^1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

Where, μ = refractive index of lens

μ^1 = refractive index of surrounding medium.

R_1 and R_2 are the radius of a curvature of two sides of lens.

$$\text{So } \frac{1}{F} \propto (\mu - \mu^1)$$

Hence, μ^1 for water is greater than air, so $(\mu - \mu^1)$ decreases and focal length increases in water.

43. Lamps used in street lighting are connected in

- (a) Both series and parallel
(b) Series

- (c) Parallel
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (c)

In a parallel connection, all the electrical components are arranged in parallel to one another. The amount of current that flows through each component adds up to form the current flow through the source. In a parallel connection, when the resistors are parallel to each other, the amount of voltage that flows through each resistor is the same.

Lamps used for street lighting are connected in parallel because all the lamps have same voltage.

Heat and Thermodynamics

1. With the increase of pressure, the boiling point of any substance

- (a) increases (b) decreases
(c) remains same (d) becomes zero
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

Boiling is the process by which a liquid turns into a vapor when it is heated to its boiling point. The change from a liquid phase to a gaseous phase occurs when the vapor pressure of the liquid is equal to the atmospheric pressure exerted on the liquid. Boiling is a physical change and molecules are not chemically altered during the process. When we increase the atmospheric pressure above the solution, the boiling point of a solution increases.

2. The temperature point at which the solid, liquid and gaseous states may stay together is known as

- (a) Boiling point (b) Melting point
(c) Freezing point (d) Triple point
(e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

The temperature point at which the solid, liquid and gaseous states may stay together is known as the triple point. In thermodynamics, a substance's triple point is the unique combination of temperature and pressure at which the solid phase, liquid phase and gaseous phase can all coexist in thermodynamic equilibrium. For example, at 611.657 Pascal partial air pressure, 273.16K (0.01°C) is the triple point of water at which liquid water, solid ice and water vapor can be found together.

3. Which of the following has highest boiling point?

- (a) H_2O (b) H_2S
 (c) H_2Se (d) H_2Te
 (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (a)

In the given options the highest boiling point is H_2O . The boiling point of water is 100 degree Celsius.

4. Water is an excellent cooling agent because of its

- (a) low boiling point
 (b) high boiling point
 (c) high specific heat
 (d) low specific heat
 (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

Water has high specific heat. Water neither heats up nor cools down quickly. This property makes water an excellent cooling agent. Water needs a large amount of heat to become hot, and it needs to lose a large amount of heat to become cold. In other words, water neither heats up nor cools down quickly.

5. At which temperature, the Celsius and Fahrenheit scales coincide?

- (a) 0°C (b) 10°C
 (c) -40°C (d) 40°C
 (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

-40° is the temperature, at which the readings are same in both degrees Centigrade and in Fahrenheit scale. Relation between degree Centigrade & degree Fahrenheit is

$$\frac{^\circ\text{C}}{5} = \frac{^\circ\text{F} - 32}{9}$$

Placing $^\circ\text{F} = -40$ to verify

$$^\circ\text{C} = \frac{(-40 - 32) \times 5}{9} = \frac{-72 \times 5}{9} = -40^\circ\text{C}$$

6. Normal temperature of human body on Celsius scale will be –

- (a) 310 degree (b) 98.4 degree
 (c) 36.9 degree (d) 31.5 degree

41st B.P.S.C. (Pre) 1996

Ans. (c)

The formula to convert Fahrenheit ($^\circ\text{F}$) to Celsius ($^\circ\text{C}$) is

$$^\circ\text{C} = \frac{(^{\circ}\text{F} - 32) \times 5}{9}$$

Since the temperature in $^\circ\text{F} = 98.4^\circ$

$$\therefore ^\circ\text{C} = \frac{(98.4 - 32) \times 5}{9} = 36.89^\circ\text{C}$$

7. The value of 40 degree Celsius in Fahrenheit scale is :

- (a) 104°F (b) 100°F
 (c) 102°F (d) 75°F
 (e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

$$^\circ\text{F} = \frac{9}{5} ^\circ\text{C} + 32$$

where, C = Centigrade Temperature

$^\circ\text{F}$ = Fahrenheit Temperature

$$^\circ\text{F} = \frac{9}{5} \times 40 + 32 \Rightarrow ^\circ\text{F} = 72 + 32 \Rightarrow ^\circ\text{F} = 104$$

So, the value of 40°C is 104°F at Fahrenheit scale.

8. The value of 50°C in Fahrenheit scale is :

- (a) 104°F (b) 122°F
 (c) 100°F (d) 75°F
 (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

$$^\circ\text{F} = \frac{9}{5} ^\circ\text{C} + 32$$

$$\Rightarrow ^\circ\text{F} = \frac{9}{5} \times 50 + 32$$

$$\Rightarrow ^\circ\text{F} = 90 + 32 \Rightarrow ^\circ\text{F} = 122$$

So, the value of 50°C is 122°F at Fahrenheit scale.

9. What is the value of absolute zero on the Fahrenheit scale?

- (a) 0°F
 (b) -22°F
 (c) -459.4°F
 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

Absolute zero is the lowest possible temperature where heat energy is absent in a substance.

value of Absolute zero = $-273^{\circ}\text{C} = 0$ Kelvin

$$\frac{{}^{\circ}\text{C} - 0}{100} = \frac{{}^{\circ}\text{F} - 32}{180}$$

$$= \frac{\text{K} - 273}{100}$$

$$\frac{-273}{100} = \frac{{}^{\circ}\text{F} - 32}{180}$$

$$-273 \times 1.8 = {}^{\circ}\text{F} - 32$$

$${}^{\circ}\text{F} = -491.4 + 32$$

$${}^{\circ}\text{F} = -459.4$$

Hence the value of absolute zero on the Fahrenheit scale is -459.4°F

10. In winter when you touch a log of wood and a rod of iron, you will feel that rod of iron is colder than wood because –

- (a) The heat of rod of iron is less than the heat of log of wood
- (b) In comparison to wood, iron is good conductor of heat
- (c) In comparison to wood, iron is bad conductor of heat
- (d) The rod of iron is heavier than the log of wood

41st B.P.S.C. (Pre) 1996

Ans. (b)

In winter, if we touch a log of wood and a rod of iron, we feel that rod of iron is cold because in comparison to wood, iron is a good conductor of heat. When both the metal and the wood are at the same temperature, heat will flow from iron. Similarly, when we touch wood, a poor conductor, heat does not flow from it. Hence iron rod will feel colder as compared to wood.

11. Which allotrope of carbon is a good conductor of electricity?

- (a) Coke
- (b) Diamond
- (c) Graphite
- (d) Fullerene

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

Graphite, allotrope of carbon is a good conductor of electricity. Allotropes are two or more forms of the same elements that exist in the same physical state but have physical and chemical properties that differ from one another. Carbon exhibits many allotropes. Diamond and Graphite are two well-known crystalline forms of carbon. Diamond is the hardest substance on the earth and in graphite, electrons are delocalized over the whole sheet so it conducts electricity along the sheet.

12. Food gets cooked faster in a pressure cooker because –

- (a) water starts boiling at low temperature due to high pressure
- (b) water starts boiling at high temperature due to high pressure
- (c) water starts boiling at low temperature due to low pressure
- (d) water starts boiling at high temperature due to low pressure
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Food can be cooked faster in pressure cooker since the boiling point of water increases with increase in pressure. As the water boils, the steam produced cannot evaporate so the pressure inside the cooker increases. The boiling point of water now becomes about 120°C . The load pin fixed on the heavy aluminium lid prevents any escape of steam. The rubber ring around the lid seals the liquid inside. Thus food gets cooked faster in pressure cooker.

13. The working principle of a washing machine is :

- (a) reverse osmosis
- (b) diffusion
- (c) centrifugation
- (d) dialysis
- (e) None of the above / More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

The washing machine works on the theory of centrifugation. The reaction force that is equal in magnitude to the centripetal force but whose direction is opposite to the centripetal force (i.e. outwards from the centre) is called centrifugal force. The machine used for cloth cleaning, used for butter extraction from milk etc .

14. The process of separating different dissolved constituents of a mixture by their adsorption on an appropriate material is called

- (a) diffusion
- (b) distillation
- (c) chromatography
- (d) centrifugation
- (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

chromatography is a technique used for the separation and identification of dissolved constituents of a mixture by adsorbing them over an appropriate material.